Agent Verifiers in Counterfactual Analysis: A Pilot Study Benchmarking Agentic Heterogeneous Treatment Effects

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Abstract

Language-model agents can expand the covariate space of randomized experiments by retrieving and summarizing the fast-moving local context that static baselines miss, with the goal of estimating more precisely who benefits most from an intervention. This capability creates two challenges to verification. The first is *retrieval leakage*, in which post-treatment information enters nominally pretreatment covariates. The second is *encoder (weight-based) leakage*, in which a pretrained encoder injects knowledge acquired after treatment into the representation even when its input text is clean, i.e., has passed temporal gatekeeping and contains no post-treatment content. We develop a three-layer benchmark protocol for examining whether agentic treatment-effect representations are admissible before they enter causal analysis: (i) *retrieval admissibility*, enforced by temporal gatekeeping and formalized by a two-channel information bound that limits the contribution of event-referencing pretraining records while leaving any residual summary channel explicit; (ii) *heterogeneity validity*, scored by rank-weighted average treatment effect (RATE) ratios computed from cross-fitted priority scores under prespecified rules separating descriptive from confirmatory use; and (iii) *encoder admissibility*, verified with chronologically vintaged encoders, future-fact probes, and paired known-leakage controls. As a pilot, we apply the protocol to the 2008 Uganda Youth Opportunities Program randomized trial, a ground-truth environment in which agent-produced data can be examined against known outcomes and treatment assignments: temporally gatekept agentic text yields a consistent descriptive ordering over researcher-collected tabular covariates across all five outcomes examined, and vintaged-encoder probes detect no clear weight-based leakage under the settings examined. The contribution is the verification methodology itself—a first step toward a reusable framework for examining whether an agent-produced representation is causally admissible and genuinely encodes heterogeneity, in a setting where counterfactual outcomes are never observed—together with an explicit accounting of what the benchmark does and does not establish.

1 Introduction

Randomized controlled trials (RCTs) estimate average treatment effects, but practical decisions—whom to target, where to expand, when to redesign—depend on identifying who benefits most. The individual effect $Y_i(1) - Y_i(0)$ is not jointly observable; the prediction target in conditional average treatment effect (CATE) estimation is instead $\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$ article

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Theorem Lemma Corollary Assumption Remark

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2 Introduction

Randomized controlled trials (RCTs) estimate average treatment effects, but practical decisions—whom to target, where to expand, when to redesign—depend on identifying who benefits most. The individual effect $Y_i(1) - Y_i(0)$ is not jointly observable; the prediction target in conditional average treatment effect (CATE) estimation is instead $\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$ [1–4]. The quality of any CATE estimate is bounded by the quality of the pretreatment covariates it conditions on. Researchers rarely know in advance which signals matter most.

Recent work has expanded the covariate space of experiments with new data modalities: satellite imagery [5, 6], text [7], and vision-language representations [8]. Web-based text is of particular interest because it uniquely records the fast-moving local context—unrest, price shocks, infrastructure projects, the political climate surrounding a rollout—that slow-changing baselines miss. Language-model agents are a natural tool for retrieving such text at scale. But this raises a verification question: when an agent retrieves and summarizes context on our behalf, does the resulting analysis actually get better? Verification is well understood only where ground truth is clear. Heterogeneous causal inference is doubly difficult: the estimand itself has no per-unit ground truth, and the agent's outputs—retrieved documents, summaries, embeddings—are high-dimensional and only loosely specified. How to verify agent-produced representations in this counterfactual regime is, to our knowledge, an open question; this paper is a pilot investigation of what such verification requires and where it breaks down. Most agent evaluation benchmarks focus on task success, instruction following, tool-use accuracy,

or interaction quality in interactive environments [9]; holistic evaluation goes further and argues for composing multiple capability signals [10]. Here we address a different verification problem: whether an agent's *data-producing* behavior yields representations that are causally admissible for downstream treatment-effect estimation. An agent that retrieves and summarizes competently can still silently bias a causal estimate if its outputs violate the pretreatment constraint. This is a data-admissibility problem. Because verification signals degrade under selection pressure [11], the checks must be prespecified.

Causal inference is a sharp setting for verification. The target—a covariate's contribution to treatment-effect heterogeneity—cannot be checked against ground truth directly, because individual treatment effects are unobservable and randomization provides only indirect anchors. Errors are also silent: a leakage-corrupted representation need not degrade any visible task metric, yet it can bias who is deemed to benefit. A protocol that works here, where failures are invisible by design, sets a high bar for agent-produced data more broadly.

An agent's output can corrupt causal inference through two distinct channels. First, *retrieval or summary leakage*: unstructured retrieval easily pulls post-treatment information into nominally pretreatment covariates, and an archival timestamp establishes only that a version existed by the cutoff, not its locality, factual accuracy, or revision history. Second, *weight-based (encoder) leakage*: pretrained encoders carry posterior knowledge from massive post-treatment corpora and can project anachronistic information into embeddings even when the input text passes temporal gates. Restricting data collection is therefore necessary but insufficient and the two channels must be verified separately.

We propose a benchmark protocol for *agentic heterogeneous treatment effects* with three verification layers. **Layer 1 (retrieval admissibility)**: an agent operationalizes the pretreatment filtration as a strict temporal gatekeeper, paired with a two-channel information bound (Theorem 1) that limits the contribution of event-referencing pretraining records while leaving any residual summary channel explicit. **Layer 2 (heterogeneity validity)**: rank-weighted average treatment effect (RATE) ratios [12] computed from cross-fitted priority scores, with prespecified rules separating descriptive from confirmatory use. **Layer 3 (encoder admissibility)**: chronologically vintaged encoders [13], future-fact probes, paired known-leakage controls, and synthetic leakage injections. We instantiate the protocol on the 2008 Uganda Youth Opportunities Program (YOP) RCT [14, 15]—a rare ground-truth environment in which treatment assignment is known, and outcomes are observed, so the agent's contribution can be audited. We are explicit about scope: the contribution is the verification methodology; the empirical results illustrate the protocol in a single pilot setting, and the headline comparison is descriptive.

In the case study, temporally gatekept agentic text attains higher descriptive fold-averaged RATE ratios than researcher-collected covariates on all five outcomes examined, although none exceeds the 1.96 reference value and the current aggregation does not justify that threshold. Vintaged-encoder probes, calibrated by known-leakage controls, detect no robust weight-based leakage under the settings examined. Section 6 states precisely what the benchmark does and does not establish.

Our contributions are as follows. First, we formalize agentic treatment-effect representation as a verification problem with two distinct leakage channels: retrieval/summary leakage and encoder weight-based leakage. Second, we propose a three-layer benchmark protocol consisting of retrieval admissibility, heterogeneity validity, and encoder admissibility. Third, we pilot the protocol on the Uganda Youth Opportunities Program RCT, using it as a ground-truth environment for examining agent-produced representations. Fourth, we provide an explicit accounting of what the benchmark does and does not establish, including the separation between descriptive heterogeneity ordering and confirmatory causal claims.

3 Background and problem setup

Admissible agentic representations. For unit i, let t_i denote the intervention date and X_i the raw covariates; the CATE over raw covariates is $\tau_{\mathcal{X}}(x) = \mathbb{E}[\, Y_i(1) - Y_i(0) \mid X_i = x \,]$. A representation is *admissible* only if it can be constructed exclusively from information available before t_i; we write $\mathcal{F}_{i,t_i^-}$ for this verifiably pre-t_i information set. Let $\mathcal{L}_i$ denote the study-defined locality of unit i. For each document d, let $\mathrm{loc}(d)$ denote its locality and $\mathrm{pubtime}(d)$ its publication or archival timestamp.

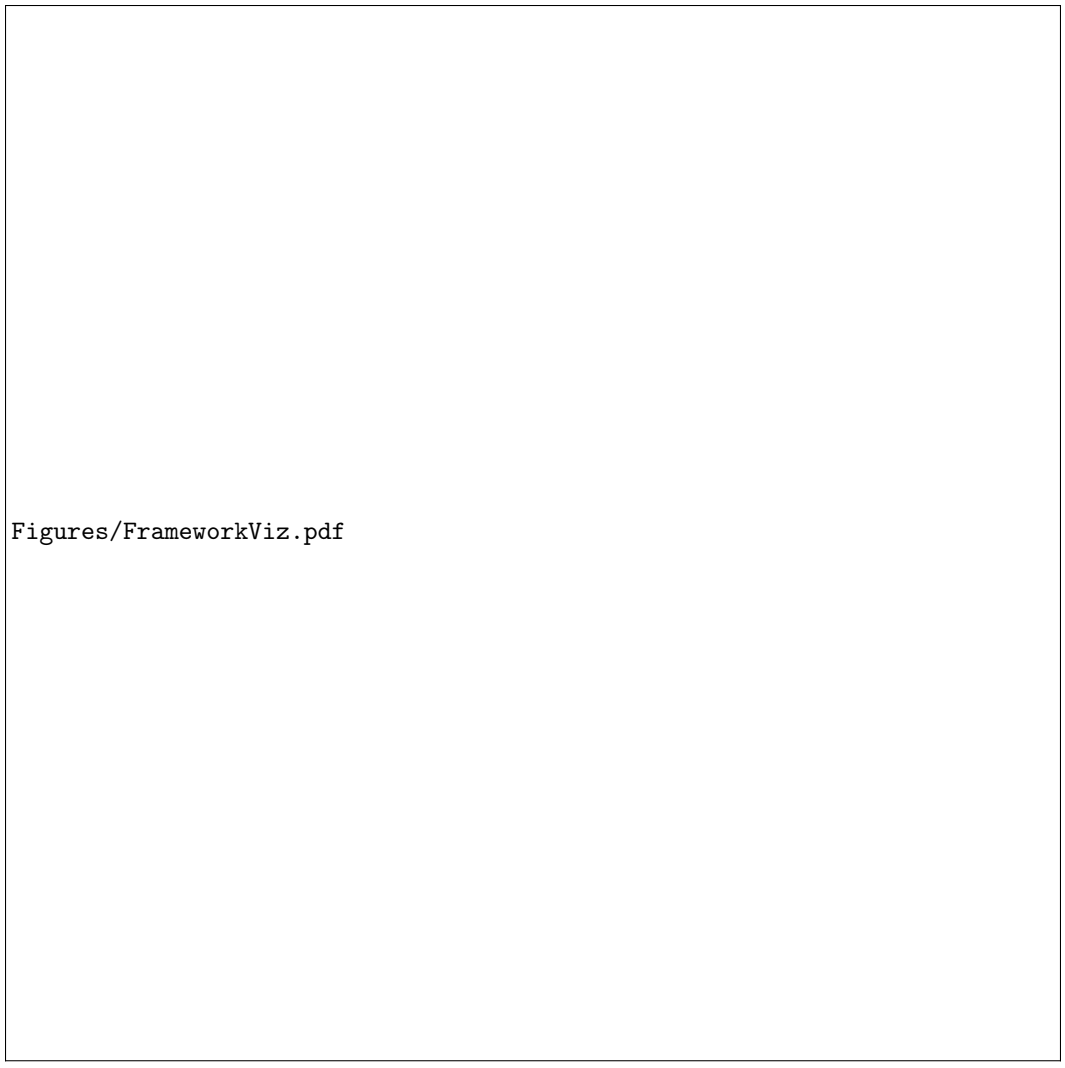


Figure 1: Key challenges in verifying agentic workflows in causal systems.

The eligible historical corpus is

$$\mathcal{D}_{i,t_i^-} = \big\{\, d : \mathrm{loc}(d) \in \mathcal{L}_i,\ \mathrm{pubtime}(d) < t_i \,\big\}. \tag{1}$$

Archival verification of $\mathrm{pubtime}(d)$ establishes only that the verified version existed before t_i; locality, factual accuracy, and revision history require separate checks. Let Θ_{π_i} collect every parameter or model weight used by the retrieval-and-summarization policy. The policy is non-anticipating only when all of its inputs are admissible and

$$S_i = \pi_i\Big(\mathcal{D}_{i,t_i^-},\, \Theta_{\pi_i},\, \epsilon_i\Big), \qquad \epsilon_i \perp\!\!\!\perp \Big(Y_i(0),\, Y_i(1),\, X_i,\, t_i,\, \mathcal{D}_{i,t_i^-},\, \Theta_{\pi_i}\Big). \tag{2}$$

This is a measurability or non-anticipation condition, not statistical independence from a future outcome: useful pretreatment text may predict that outcome, and post-period information may enter through Θ_{π_i} if it is not separately controlled. Encoding the summary with a numerical text encoder g_ϕ induces a learned-representation estimand space,

$$Z_i = g_\phi(S_i), \qquad \tau_{\mathcal{Z}}(z) = \mathbb{E}[\,Y_i(1) - Y_i(0) \mid Z_i = z\,]. \tag{3}$$

Two admissibility conditions must be verified separately. *Retrieval admissibility* requires that no document with unverifiable timing enter S_i. *Encoder admissibility* requires that g_ϕ itself not inject knowledge learned after t_i (weight-based leakage). The benchmark attaches a distinct verification layer to each (Fig. 2).

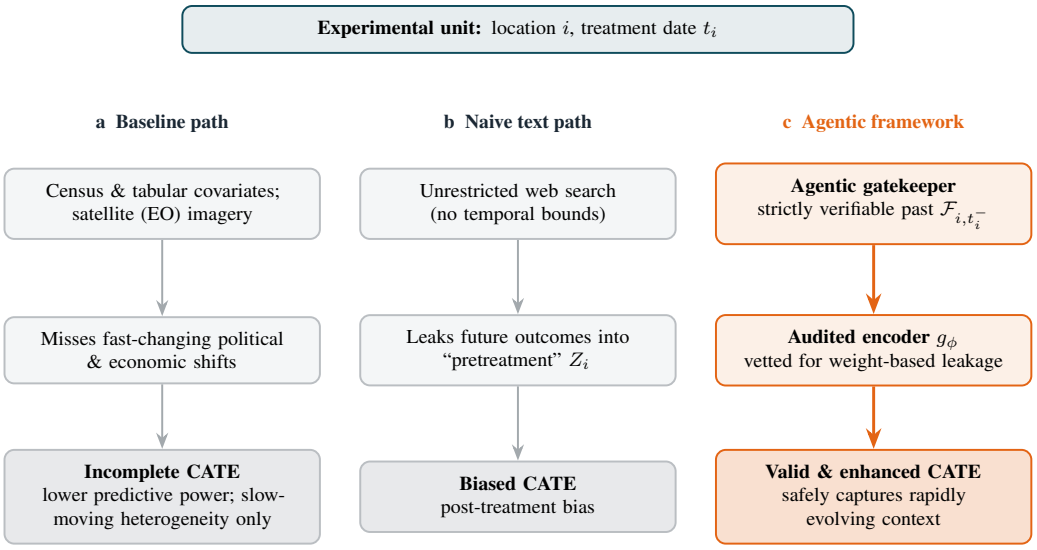


Figure 2: **Why agent-produced representations need verification.** **a**, Baseline representations (census covariates, Earth-observation imagery) are admissible but miss fast-changing political and economic context. **b**, Naive web retrieval captures that context but leaks future outcomes into nominally pretreatment representations. **c**, The agentic framework retains the contextual signal while checking temporal provenance at retrieval (Layer 1) and checking the encoder g_ϕ (Layer 3); Layer 2 then scores heterogeneity using the RATE protocol.

The RATE ratio as a verification metric. Because CATEs have no per-unit ground truth, the benchmark verifies heterogeneity with a rank-based functional. Given out-of-fold priority scores from a CATE estimator, the rank-weighted average treatment effect (RATE) weights units by how strongly they are prioritized, with AUTOC weighting and Qini weighting as a robustness check [12]. The RATE ratio is

$$\text{RATE ratio} = \widehat{\text{RATE}} \,/\, \widehat{\text{se}}\big(\widehat{\text{RATE}}\big). \tag{4}$$

For one prespecified RATE estimator satisfying the cited regularity conditions, the centered statistic $(\widehat{\text{RATE}} - \text{RATE})/\widehat{\text{se}}(\widehat{\text{RATE}})$ is asymptotically standard normal; the uncentered ratio has that null interpretation only under $H_0 : \text{RATE} = 0$. The metric answers the verification question *does this representation separate beneficiaries from non-beneficiaries, out of sample?* without requiring ground-truth individual effects.

4 The benchmark protocol: three verification layers

4.1 Layer 1: retrieval admissibility

Operationalization. For each located, dated experimental unit, the agent searches the open web for *local* documents capturing the unit's context; verifies place tags and publication timestamps strictly prior to t_i; consults Internet Archive Wayback Machine snapshots where needed. Two filters operationalize admissibility at the representation stage: *no-post* drops any document whose verified timestamp is not strictly before t_i; *known-pre* additionally requires affirmative archival evidence of pre-t_i existence. The protocol treats agent failures—misread sources, imperfectly honored date restrictions—as retrieval-admissibility violations for these filters to catch.

Formal guarantee: a two-channel bound. Temporal filtering controls which documents enter a summary but does not imply that the summary is statistically independent of a later fact. We therefore bound the training-data contribution and retain the summary channel explicitly. Let Ψ denote a post-t_i quantity correlated with the potential outcomes—the realized outcome or, in a diagnostic probe, a known future label. The encoder weights are produced by a pretraining algorithm

$$\phi \;=\; \mathcal{A}(\mathcal{C}_{t_i^-}, \mathcal{G}, R),$$

where $\mathcal{C}_{t_i^-}$ is the pre-t_i portion of the pretraining corpus, $\mathcal{G} = (g_1, \ldots, g_k)$ consists of k post-t_i training records referencing the event, and R is the algorithm's internal randomness. This is a closed structural model: every data source or selection rule through which Ψ can affect training must appear among $(\mathcal{C}_{t_i^-}, \mathcal{G}, R)$. Three assumptions—representation construction, a closed training model, and per-record KL stability with constant β—are stated in Appendix A.

Theorem 1 (Two-channel information bound)**.** *Under Assumptions 1–3,*

$$\mathrm{I}(Z_i; \Psi \mid \mathcal{C}_{t_i^-}) \;\le\; k\beta + \mathrm{I}(S_i; \Psi \mid \phi, \mathcal{C}_{t_i^-}).$$

Moreover, $\mathrm{I}(\phi; \Psi \mid \mathcal{C}_{t_i^-}) \le \mathrm{I}(\phi; \mathcal{G} \mid \mathcal{C}_{t_i^-}) \le k\beta$*. For* $k = 0$*, the product* $k\beta$ *is interpreted as zero.*

The contribution of event-referencing pretraining records is bounded by their number k and per-record KL influence β, while any residual information carried by the selected summary remains explicit. Temporal document filtering alone does not zero the summary term; separating legitimate predictive pretreatment content from leakage through Θ_{π_i} requires a baseline of admissible pretreatment information and an explicit conditional-independence model. Because β is not estimated for the encoders examined, Theorem 1 is a structural guarantee, not a numerical leakage bound. Proofs, a differential-privacy corollary, and scope remarks appear in Appendix A.

4.2 Layer 2: heterogeneity validity

Protocol. For each candidate representation—tabular covariates X_i, agentic text Z_i, and concatenations such as (X_i, Z_i)—a causal forest [2, 16] is trained with identical folds, samples, and hyperparameters; out-of-fold priority scores are formed; and the AUTOC-weighted RATE ratio (4) is computed with a cluster-robust standard error at the randomization-unit level (Qini weighting as robustness).

Discipline. What makes this layer a verification instrument is the prespecified separation of descriptive and confirmatory use: (i) averages of fold-specific ratios are *descriptive*—without the joint fold covariance, the average does not inherit a standard-normal null, so a 1.96 cutoff is not valid for it; (ii) confirmatory inference forms one out-of-fold priority score per observation and one cluster-robust RATE estimate on the full evaluation sample, prespecified before results are inspected; (iii) comparing representations requires a paired, covariance-aware contrast of the two priority rules; and (iv) any search over web signals or specifications must occur in a nested training sample or with held-out, multiplicity-adjusted evaluation. These rules exist because verification signals degrade under selection pressure [11]: a benchmark that reports only its best-finding ratio verifies nothing.

4.3 Layer 3: encoder admissibility

Design. Temporal gatekeeping makes the *documents* pre-t_i, but it does not make the *encoder* pre-t_i. We probe this channel with chronologically vintaged encoders trained on chronologically bounded corpora [13]: hold the input text fixed and always pre-event, vary only the encoder vintage, and ask whether later vintages recover future facts from the same historical text better than earlier vintages. Two probe families are used. The *death-year probe* predicts subjects' later death years from biography snapshots written while the subjects were alive. The *earthquake probe* asks whether post-event encoder vintages move pre-event city pages for affected cities closer to earthquake concepts than matched control cities.

Calibration instruments. Every clean probe is paired with a known-leakage control that inserts the future label explicitly into the input, validating the input-to-readout path; synthetic leakage injections verify that an inserted association is detectable by the harness; and a deterministic lexical encoder provides a sanity check (all shifts exactly zero).

Epistemic scope. Intervals covering zero establish non-detection under the design, not equivalence to zero. Explicit-label controls validate the input channel but are not weight-only known-leakage controls; a weight-only control would require otherwise matched model variants with a synthetic future association inserted only during pretraining. A vintage comparison changes all intervening training data and optimization history, not only event-referencing records, and is therefore a diagnostic, not a literal intervention on $\mathcal{G}$. The benchmark reports this scope explicitly.

5 Pilot case study: a first instantiation

We now exercise the protocol on a single, well-characterized randomized setting. This is a pilot analysis: its purpose is to explore whether each layer of the benchmark is runnable in practice and yields inspectable evidence, not to establish a new substantive fact about the program. A full-scale evaluation across studies and pipelines is left for future work.

5.1 Setting

The YOP experiment randomized unconditional cash grants (approximately US$400 per member) to young-adult groups that applied to Uganda's Youth Opportunities Program in 2008 [14, 15]. Our analysis file covers 3,142 individuals in 535 group localities; estimation samples comprise roughly 1,730 individuals in 439 group clusters per outcome. We estimate intention-to-treat heterogeneity on five outcomes: log training hours, any vocational training, shift into a skilled trade, a human capital index, and an income index. The tabular baseline uses the covariates collected by the original study team, matched to the same estimation samples.

The agent is an autonomous coding agent driven by a general-purpose language model; per context it returns at most eight sourced contextual factors under a budget of 24 searches (per-query cap of 8 results) and a 1,800 s timeout (with a 900 s reduced-budget fallback). Completed contexts return a mean of 6.2 factors. Retrieved factors are embedded with a commercial embedding model (large variant primary; small variant and tf–idf as robustness checks), aggregated by max pooling across factors per context, and reduced to 32 principal components. CATE functions are estimated with causal forests (1,000–4,000 trees, honest splitting) with cluster-robust estimation at the group level. Full protocol details, including model families, are in Appendix B.

5.2 Layer 2 applied: a descriptive heterogeneity ordering

Table 1 and Fig. 3 report fold-averaged RATE ratios by outcome and representation. Agentic-text AUTOC ratios range from 0.49 (income index) to 1.78 (training hours), against tabular baselines between -0.07 and 0.22; descriptive differences range from $+0.4$ to $+1.7$. These differences are neither standardized treatment-effect gains nor covariance-aware tests of representation equality. All five text AUTOC ratios are below 1.96, and the current fold aggregation does not justify that threshold in any case. The 1.96 value is shown only as the reference critical value for a single prespecified RATE statistic; it is not a valid cutoff for the fold-averaged descriptive quantities reported here. The descriptive ordering is preserved under Qini weighting and is largest for program-proximate outcomes (training hours, training take-up, trade shift), consistent with local context—labor-market conditions, program-rollout climate—shaping take-up and training intensity more directly than long-run income. The income-index text representation is a temporally filtered tf–idf specification, selected by its outcome result. The protocol thus produces a consistent descriptive ordering: temporally gatekept agentic text is more strongly associated with recoverable treatment-effect heterogeneity than the tabular baseline across all five outcomes. Whether this pattern persists beyond this pilot setting is an open question. Under the prespecified discipline, however, this ordering is not evidence of confirmatory superiority; establishing that requires the cross-fitted, paired, covariance-aware inference prescribed in Section 4. The ordering is useful because it identifies where confirmatory evaluation should be focused.

5.3 Layer 3 applied: no robust vintage-associated signal detected

We probe chronologically vintaged ChronoGPT encoders with training cutoffs in 2007, 2014, and 2024 [13].

Death-year probe. Historical Wikipedia biography revisions were sampled for people alive at snapshot time who later died in known years (2007 snapshot $n = 76$; 2014 snapshot $n = 184$). Clean historical text does not robustly predict later death years and is identical across encoder vintages (Spearman -0.165 and 0.000; window-classification balanced accuracy 0.347 and 0.304; late-death AUC 0.404 and 0.511), while known-leakage controls with the death year inserted are detected nearly perfectly (Spearman 0.979 and 0.980; balanced accuracy and AUC 1.00; Table 2).

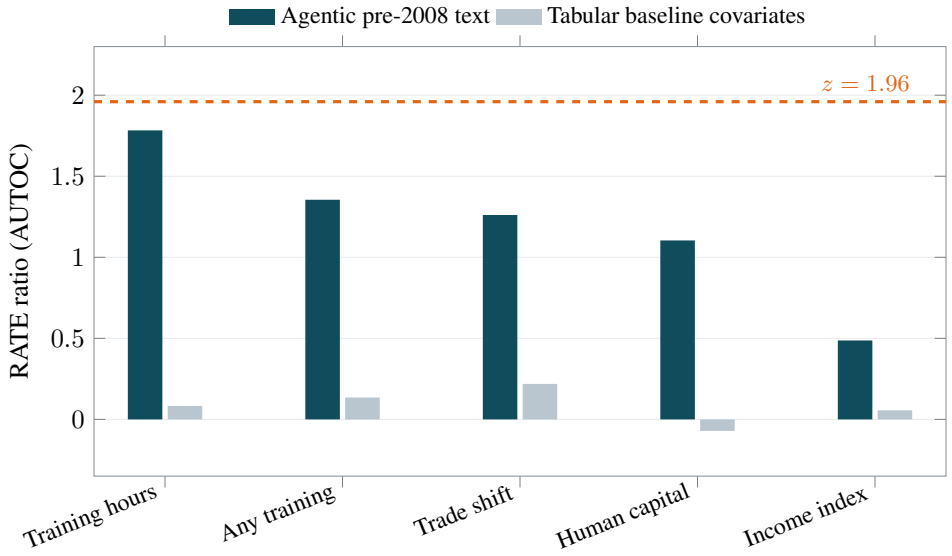


Figure 3: **Descriptive RATE ratios for agentic pre-2008 text and tabular baseline covariates in the Uganda YOP RCT.** Fold-averaged RATE ratios (AUTOC weighting) for causal forests on temporally filtered agentic text embeddings versus matched tabular baselines, for five ITT outcomes. The dashed line marks 1.96, the positive critical value in a two-sided 5% standard-normal test of H_0 : RATE $= 0$ for one prespecified RATE ratio; it is not valid for the present average of fold-specific ratios. Qini-weighted descriptive results are in Table 1.

Table 1: **Uganda YOP descriptive RATE ratios by outcome and representation.** AUTOC- and Qini-weighted RATE ratios for agentic-text and matched tabular representations, with descriptive differences (Δ, agentic text minus tabular). The differences are not paired tests or treatment-effect gains. Agentic text uses factor-level embeddings of temporally filtered documents (income index: temporally filtered tf–idf representation; see Appendix B).

	AUTOC ratio			Qini ratio		
Outcome	Agentic text	Tabular	Δ	Agentic text	Tabular	Δ
Training hours (log)	1.78	0.08	+1.70	1.82	0.59	+1.23
Any vocational training	1.36	0.14	+1.22	1.47	0.76	+0.71
Shift into skilled trade	1.26	0.22	+1.04	1.40	0.02	+1.38
Human capital index	1.10	−0.07	+1.17	1.04	−0.23	+1.27
Income index	0.49	0.06	+0.43	0.69	−0.06	+0.75

Earthquake probe. Pre-event Wikipedia city pages for 11 major earthquakes were matched to control cities outside the affected area (13 matched pairs). Post-event shifts in the affected-minus-control embedding distance gap to earthquake concepts are near zero with pair-bootstrap intervals covering zero (2024 vintage: 0.005, 95% interval $[-0.047,\ 0.058]$), and no specificity contrast against flood and generic-city concepts is detected (Fig. 4).

Sensitivity and calibration. Alternative pooling choices and regularized canonical-correlation alignment of embedding spaces can flip the sign of small aggregate shifts depending on anchor choice, and event-level results are heterogeneous; a model-identity diagnostic bounds the vintage footprint in the stored geometry (balanced accuracy 0.27–0.42 on clean text). Synthetic leakage simulations verify that injected associations are detectable (best Spearman rising from 0.44 to 0.82 at $n = 20$ and from 0.55 to 0.93 at $n = 30$ under strong injection). The defensible conclusion is that no robust vintage-associated signal is detected under the settings examined; neither non-detection nor the current controls establish that the encoders are leakage-free.

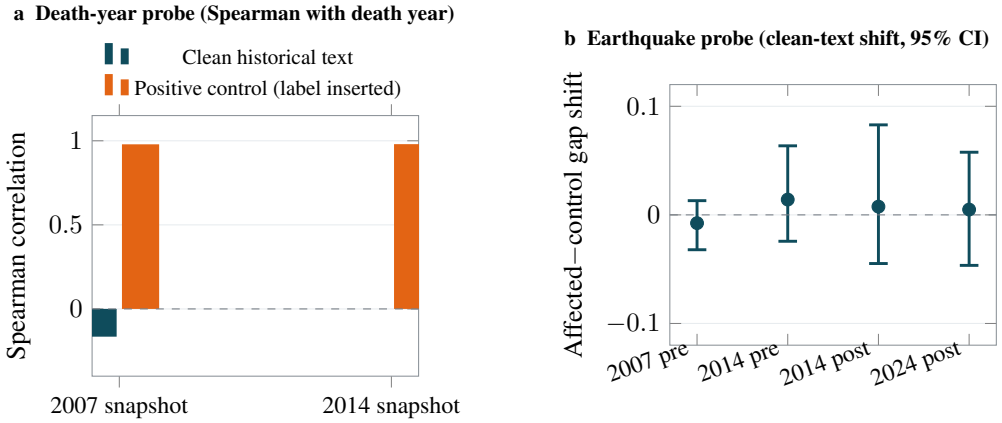


Figure 4: **Vintage comparison with chronologically trained encoders.** **a**, Death-year probe: Spearman correlation between embedding-based predictions and subjects' later death years, for clean historical biography text versus known-leakage controls with the death year inserted (2007 cohort $n = 76$; 2014 cohort $n = 184$; clean-text metrics are identical across encoder vintages). **b**, Earthquake probe: clean-text shift in the affected-minus-control embedding distance gap to earthquake concepts, by encoder vintage and event phase (mean-first-16 pooling; error bars, 95% pair-bootstrap intervals; 13 matched city pairs nested in 11 events). Leakage would appear as negative post-event shifts for later vintages.

Table 2: **Death-year probe: no robust clean-text signal is detected; explicit-input controls are nearly perfect.** Out-of-fold probe performance for predicting later death years from ChronoGPT embeddings of fixed historical biography snapshots. Rows without an inserted death-year label receive identical input text across encoder vintages (2007/2014/2024) within each revision cutoff (2007 or 2014). MAE, mean absolute error in years (categories); window bal. acc., balanced accuracy for death-window classification; AUC, area under the curve for late-death classification.

Snapshot	Condition	n	MAE	Spearman	Window bal. acc.	Late-death AUC
2007	clean historical text	76	1.196	−0.165	0.347	0.404
2007	positive control (label inserted)	76	0.017	0.979	1.000	1.000
2014	clean historical text	184	1.237	−0.000	0.304	0.511
2014	positive control (label inserted)	184	0.007	0.980	1.000	1.000

6 Discussion

What the benchmark provides. In the current instantiation, the benchmark yields four verifiable outputs: (i) documented provenance checks—every retained document carries a verified pre-t_i timestamp, and the two filters (*no-post*, *known-pre*) are reapplied at the representation stage; (ii) a structural bound on the training-record channel of encoder leakage under explicit assumptions, with the summary channel left visible (Theorem 1); (iii) calibrated encoder probes, in which known-leakage controls validate the input-to-readout path, synthetic injections validate detectability, and a deterministic encoder validates the harness; and (iv) a descriptive, metric-defined heterogeneity ordering computed under identical folds, samples, and hyperparameters, so that comparisons across representations are like-for-like.

Beyond causal inference. The three layers separate concerns that arise whenever agents produce data for downstream decisions. Temporal gatekeeping applies to any task with an information constraint on inputs—forecasting, nowcasting, or prediction settings where labels must not leak into RATE-ratio discipline—prespecified descriptive versus confirmatory use of a ranking metric—and transfers to features. Any prioritization decision is built on agent-produced signals. Vintage probes transfer wherever a pretrained model's corpus may overlap with evaluation targets, echoing contamination checks for language-model benchmarks. The protocol template in the supplementary material operationalizes this transfer.

Limitations. The pilot case study exercises the protocol on one randomized setting. The income-index text representation was selected by its outcome result, and a confirmatory run would predeclare this choice or select it within nested training data. Probe coverage is limited to particular encoder families, pooling regimes, and event classes. The sensitivity analyses also show that embedding-geometry conclusions can depend materially on pooling and alignment choices. Finally, agents can misread sources or imperfectly honor date restrictions; the protocol treats such failures as retrieval-admissibility violations for the filters to catch. Extending the protocol to additional studies, estimation pipelines, and spatiotemporal contexts is a natural next step; the present submission focuses on the verification protocol and its reference instantiation.

Positioning. This work contributes to the workshop's agenda by taking a first step toward verifying agent-produced representations used in causal inference: a pilot analysis of what such verification requires when counterfactual outcomes are never observed, and of what current designs cannot yet deliver. The framework has three components. First, *environment-grounded verification*: a randomized experiment supplies a rare ground-truth environment in which an agent's data-producing behavior can be audited directly, since treatment assignment is known and outcomes are observed. Even so, the ground truth is partial: each unit's counterfactual outcome is never observed, so verifying heterogeneity claims introduces additional methodological problems—there is no per-unit effect to check against, and the verification metric itself must be prespecified, cross-fitted, and separated into descriptive and confirmatory uses. Even so, the ground truth is partial: each unit's counterfactual outcome is never observed, so verifying heterogeneity claims introduces additional methodological problems—there is no per-unit effect to check against, and the verification metric itself must be prespecified, cross-fitted, and separated into descriptive and confirmatory uses. Second, *heterogeneous verifiable signals*: the benchmark composes a rank-based verification metric (the RATE ratio) with prespecified descriptive/confirmatory semantics. Third, *robustness and adversarial awareness*: temporal constraints are treated as properties to be checked, not trusted, because agents can fail to honor them or can effectively game the pretreatment constraint. Together, these choices make the protocol useful for verifying agent-assisted causal inference, complementing benchmarks that focus on task completion.

Availability. The encoder-probe suite—probe code, probe datasets, and results—will be released publicly. Tabular baselines can be recomputed from publicly archived trial data [15] using standard rate-ratio estimation (causal forests plus rank-weighted evaluation).

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A Formal bound: assumptions, proofs, and scope

All random elements are assumed to take values in standard Borel spaces, so regular conditional laws exist. Conditional statements are understood for $P_{\mathcal{C}_{t_i^-}}$-almost every corpus c, and mutual information and KL divergence use natural logarithms. Let Ψ denote a post-t_i quantity or fact correlated with the potential outcomes. It is a value or label, not a document—the realized outcome or, in a diagnostic probe, a known future label such as a death year or the location of a subsequent earthquake. Temporal gatekeeping may exclude future documents, but Ψ can remain predictable from past text or enter through either the summary policy's parameters or the encoder's weights.

Assumption 1 (Representation construction)**.** $Z_i = g_\phi(S_i)$ is a measurable deterministic function of (ϕ, S_i). No independence between S_i, ϕ, and Ψ is imposed.

Assumption 2 (Closed training model)**.** $\phi = \mathcal{A}(\mathcal{C}_{t_i^-}, \mathcal{G}, R)$ and $R \perp\!\!\!\perp \Psi \mid (\mathcal{C}_{t_i^-}, \mathcal{G})$. In particular, no omitted training input or selection rule may depend on Ψ once $(\mathcal{C}_{t_i^-}, \mathcal{G})$ is fixed.

Assumption 3 (Per-record KL stability)**.** The records $g_1, \ldots, g_k$ are mutually independent conditional on $\mathcal{C}_{t_i^-}$, and there exists $\beta < \infty$ such that, for $P_{\mathcal{C}_{t_i^-}}$-almost every c, every admissible realization $\mathcal{G}$, every index $j \le k$, and every admissible replacement record g',

$$D\Big(P(\phi \mid c, \mathcal{G}) \,\Big\|\, P(\phi \mid c, \mathcal{G}^{(j \to g')})\Big) \;\le\; \beta,$$

where $\mathcal{G}^{(j \to g')}$ denotes $\mathcal{G}$ with its j-th record replaced by g'.

The constant β is a uniform upper bound on the relative-entropy influence of one training record. Conditional independence of event-referencing records is a strong data-generating assumption, not a consequence of timestamping or privacy. Likewise, a finite β is not automatic: deterministic training can produce disjoint point masses and infinite KL. We do not estimate β for the encoders examined, so Theorem 1 is not a numerical leakage bound for them.

Lemma 1 (Conditional data processing)**.** *Under Assumption 1,*

$$\mathrm{I}(Z_i; \Psi \mid \mathcal{C}_{t_i^-}) \le \mathrm{I}(\phi, S_i; \Psi \mid \mathcal{C}_{t_i^-}) = \mathrm{I}(\phi; \Psi \mid \mathcal{C}_{t_i^-}) + \mathrm{I}(S_i; \Psi \mid \phi, \mathcal{C}_{t_i^-}).$$

The final term is the open summary/retrieval channel. It may contain ordinary predictive pretreatment signal as well as information introduced by a post-period retrieval or summarization policy.

Lemma 2 (Fact information is bounded by group information)**.** *Under Assumption 2,* $\mathrm{I}(\phi; \Psi \mid \mathcal{C}_{t_i^-}) \le \mathrm{I}(\phi; \mathcal{G} \mid \mathcal{C}_{t_i^-})$.

Lemma 3 (Group information under KL stability)**.** *Under Assumption 3,* $\mathrm{I}(\phi; \mathcal{G} \mid \mathcal{C}_{t_i^-}) \le k\beta$.

Proof of Lemma 1. Conditional on $\mathcal{C}_{t_i^-}$, $Z_i = g_\phi(S_i)$ is a deterministic measurable function of (ϕ, S_i). Conditional data processing and the chain rule give

$$\mathrm{I}(Z_i; \Psi \mid \mathcal{C}_{t_i^-}) \le \mathrm{I}(\phi, S_i; \Psi \mid \mathcal{C}_{t_i^-}) = \mathrm{I}(\phi; \Psi \mid \mathcal{C}_{t_i^-}) + \mathrm{I}(S_i; \Psi \mid \phi, \mathcal{C}_{t_i^-}).$$

No independence is used. □

Proof of Lemma 2. If $\mathrm{I}(\phi; \mathcal{G} \mid \mathcal{C}_{t_i^-}) = \infty$, the claim is immediate. Otherwise, for $P_{\mathcal{C}_{t_i^-}}$-almost every c the first term below is finite, and we may expand $\mathrm{I}(\phi; \Psi, \mathcal{G} \mid c)$ by the chain rule in two ways:

$$\mathrm{I}(\phi; \mathcal{G} \mid c) + \mathrm{I}(\phi; \Psi \mid \mathcal{G}, c) \;=\; \mathrm{I}(\phi; \Psi \mid c) + \mathrm{I}(\phi; \mathcal{G} \mid \Psi, c).$$

Assumption 2 gives $\phi \perp\!\!\!\perp \Psi \mid (\mathcal{C}_{t_i^-}, \mathcal{G})$, so the second term on the left vanishes. The second term on the right is nonnegative. Rearranging, $\mathrm{I}(\phi; \Psi \mid c) \le \mathrm{I}(\phi; \mathcal{G} \mid c)$. Average over c. □

Proof of Lemma 3. If $k = 0$, both sides are zero. Otherwise, fix a value $\mathcal{C}_{t_i^-} = c$ outside the exceptional null set and suppress it. By the chain rule, $\mathrm{I}(\phi; \mathcal{G}) = \sum_{j=1}^{k} \mathrm{I}(\phi; g_j \mid g_{<j})$ with $g_{<j} =$

$(g_1, \ldots, g_{j-1})$. Fix j and condition on $g_{<j}$. Writing $Q_g := P(\phi \mid g_{<j}, g)$ and taking g' as an independent copy of g_j, mutual independence of the coordinates gives $P(\phi \mid g_{<j}) = \mathbb{E}_{g'} Q_{g'}$, so

$$\mathrm{I}(\phi; g_j \mid g_{<j}) \;=\; \mathbb{E}_{g_j} D\big(Q_{g_j} \,\big\|\, \mathbb{E}_{g'} Q_{g'}\big)\,,$$

by convexity of relative entropy in its second argument and Jensen's inequality. Each Q_g is the marginal of $P(\phi \mid \mathcal{G})$ over $g_{>j}$, and both Q_{g_j} and $Q_{g'}$ are mixtures over $g_{>j}$ with the same mixing law, again by independence. Joint convexity of relative entropy therefore gives

$$D\big(Q_{g_j} \,\big\|\, Q_{g'}\big) \;\le\; \mathbb{E}_{g_{>j}} D\Big(P(\phi \mid \mathcal{G}) \,\Big\|\, P(\phi \mid \mathcal{G}^{(j \to g')})\Big) \;\le\; \beta$$

by Assumption 3. Each of the k summands is thus at most β. Average over c. □

Proof of Theorem 1. Apply Lemma 1 and bound its first summand using Lemmas 2 and 3. The displayed subsidiary chain is exactly Lemmas 2 and 3. □

The encoder-only bound $\mathrm{I}(\phi; \Psi \mid \mathcal{C}_{t_i^-}) \le k\beta$ follows only under the additional condition $\mathrm{I}(S_i; \Psi \mid \phi, \mathcal{C}_{t_i^-}) = 0$, equivalently $S_i \perp\!\!\!\perp \Psi \mid (\phi, \mathcal{C}_{t_i^-})$ when regular conditional laws are used. Temporal document filtering alone does not imply this condition.

Corollary 1 (Differential privacy as a special case)**.** *Under Assumptions 1 and 2, suppose the conditional-independence clause of Assumption 3 holds and $\mathcal{A}$ is pure ε_0-differentially private under bounded (replacement) adjacency of one training record while $\mathcal{C}_{t_i^-}$ is held fixed. Then the KL-stability clause holds with the valid, not necessarily sharp, choice $\beta = \varepsilon_0(e^{\varepsilon_0} - 1)$, and*

$$\mathrm{I}(Z_i; \Psi \mid \mathcal{C}_{t_i^-}) \le k\,\varepsilon_0\big(e^{\varepsilon_0} - 1\big) + \mathrm{I}(S_i; \Psi \mid \phi, \mathcal{C}_{t_i^-}).$$

Proof. For the output laws P, Q on replacement-neighboring datasets, pure differential privacy in both directions gives $e^{-\varepsilon_0} \le dP/dQ \le e^{\varepsilon_0}$. Hence

$$D(P\|Q) \le D(P\|Q) + D(Q\|P) = \int (dP - dQ) \log\left(\frac{dP}{dQ}\right) \le \varepsilon_0(e^{\varepsilon_0} - 1).$$

The last inequality uses $|\log(dP/dQ)| \le \varepsilon_0$ and $\int |dP - dQ| \le e^{\varepsilon_0} - 1$. Thus privacy supplies the KL clause but not conditional independence of the records; Theorem 1 supplies the conclusion. Approximate privacy is not covered by this likelihood-ratio argument, and add/remove adjacency would require the corresponding replacement or group-privacy constants. □

Remark 1 (Why condition on $\mathcal{C}_{t_i^-}$). The quantity $\mathrm{I}(Z_i; \Psi \mid \mathcal{C}_{t_i^-})$ measures residual association after conditioning on the chosen corpus. When the relevant information is finite,

$$\mathrm{I}(Z_i; \Psi \mid \mathcal{C}_{t_i^-}) = \mathrm{I}((Z_i, \mathcal{C}_{t_i^-}); \Psi) - \mathrm{I}(\mathcal{C}_{t_i^-}; \Psi).$$

It is not generally $\mathrm{I}(Z_i; \Psi)$ minus a legitimate component, and conditioning can increase mutual information. Interpreting the residual as leakage therefore requires a substantive sufficiency or causal model for $\mathcal{C}_{t_i^-}$; information theory alone does not identify temporal provenance.

Remark 2 (The retrieval channel remains open). Theorem 1 retains the retrieval and summarization channel as $\mathrm{I}(S_i; \Psi \mid \phi, \mathcal{C}_{t_i^-})$. The term can mix legitimate predictive pretreatment content with leakage through Θ_{π_i}, and timestamps alone do not separate them. Such a separation would require a baseline B containing the admissible pretreatment information and an explicit causal or conditional-independence model relative to B. If the policy is itself trained, an analogous stability analysis may bound the contribution of its weights; it cannot simply be assumed to vanish.

B Agent and estimation details

Uganda YOP data. We geocoded group localities to parish-level coordinates. Tabular baselines use the covariates collected by the original study team, matched to the same estimation samples; for example, the income-index estimation sample comprises $n = 1{,}728$ individuals in 439 clusters.

Agentic retrieval under temporal constraints. Retrieval is executed by an autonomous coding agent driven by a general-purpose language model (GPT-5.4-mini). For each context, the agent receives the locality description and treatment date and is instructed to gather local, pretreatment context relevant to treatment-effect heterogeneity. Constraints per context: at most 8 contextual factors (each a sourced paragraph with URL and publication date), a search budget of 24 queries, per-query result caps of 8, and a task timeout of 1,800 s (with a 900 s reduced-budget fallback). Publication timing is verified from page metadata and, where needed, Internet Archive Wayback Machine snapshots. This check establishes availability of a version by the cutoff, not the accuracy or locality of its contents.

Text representations and temporal filtering. Retrieved factors and their source documents are embedded with OpenAI `text-embedding-3-large` (with `text-embedding-3-small` and tf–idf bag-of-words representations as robustness checks). Two temporal filters operationalize retrieval admissibility at the representation stage: *no-post* drops any document whose verified timestamp is not strictly before t_i; *known-pre* additionally requires affirmative archival evidence of pre-t_i existence. Factor-level embeddings are aggregated per context (max pooling across factors in the primary specification) and reduced by principal component analysis to 32 components. The income-index result in Table 1 uses the temporally filtered tf–idf representation (top 150 terms), selected because it was the strongest text specification for that outcome. A confirmatory version must predeclare this choice, select it within nested training data, or adjust for selection. Embedding PCA results for that outcome are directionally similar.

CATE estimation and evaluation. CATE functions are estimated with causal forests (`grf`) [2, 16], with cluster-robust estimation at the randomization-group level, 1,000–4,000 trees, and honest splitting. Heterogeneity detection is evaluated with rank-weighted average treatment effects [12]: units are ranked by out-of-fold priority scores. The reported summaries compute an AUTOC-weighted RATE ratio (Qini as robustness) within each of five folds and average the five ratios. That average is reported descriptively because its variance depends on the fold covariance and it is not automatically standard normal. A confirmatory analysis under this protocol constructs one out-of-fold priority score per observation and then computes one RATE estimate with a cluster-robust standard error over the full evaluation sample. Identical folds, samples, and hyperparameters improve comparability but do not make a difference in marginal ratios a test of the representation effect; that requires the paired RATE difference and its covariance-aware standard error. RATE measures a prioritization functional, not policy value directly [12, 17].

C Leakage-probe details

Encoders. The probes use ChronoGPT [13], a family of language models trained with chronologically bounded corpora. The vintages used are the chronologically consistent instruct checkpoints with cutoffs 2007-12-31, 2014-12-31, and 2024-12-31 (`manelalab/chrono-gpt-instruct-v1-{20071231, 20141231, 20241231}`). All probes hold the input text fixed and historical, varying the encoder vintage. Successive vintages nevertheless differ in all intervening training text and optimization history, not only event-referencing records; the vintage comparison is therefore a diagnostic, not a literal instantiation of Theorem 1.

Death-year probe. Historical Wikipedia biography revisions were sampled for people alive at snapshot time who later died in known years (2007 snapshot: $n = 76$; 2014 snapshot: $n = 184$). Out-of-fold probes predict death year (MAE, Spearman), death window (balanced accuracy), and late death (AUC) from vintage-specific embeddings. Known-leakage controls insert an explicit `death_year=YYYY` string; these controls test recovery from the input, not from weights alone. A model-identity diagnostic (classifying which vintage produced an embedding) bounds the vintage footprint in the stored geometry (balanced accuracy 0.27–0.42 on clean text).

Earthquake probe. For 11 major earthquakes, pre-event Wikipedia city pages for affected cities were matched to control cities outside the affected area (13 pairs; 29 city-page rows). For model m, unit u, and concept prompt c, embeddings are L2-normalized and compared with cosine distance $d_m(u, c)$; the pair-level gap is $g_m(c) = d_m(\text{affected}, c) - d_m(\text{control}, c)$, and the reported shift subtracts the pre-event mean gap for that pair and concept. Specificity subtracts the corresponding

flood/generic-city negative-control shift. The primary specification pools the first 16 hidden states with L2 normalization; sensitivity analyses vary pooling (last-16) and align embedding spaces across vintages via regularized canonical correlation analysis under two anchor regimes (experiment-text anchors, 89 components; neutral anchors, 7 components). A deterministic lexical encoder provides a harness sanity check (all shifts exactly zero), and synthetic treatment-effect simulations verify that injected leakage is detectable (best Spearman rising from 0.44 to 0.82 at $n = 20$ and from 0.55 to 0.93 at $n = 30$ under strong injected leakage). These simulations test the stipulated synthetic representation perturbation; unless the association is inserted only during otherwise matched pretraining, they are not a weights-only control.

Statistics. Earthquake-probe intervals resample 13 matched pairs, although those pairs are nested in 11 earthquakes; confirmatory uncertainty should resample or cluster at the independent event level and report small-sample sensitivity. No statistical method was used to predetermine sample sizes, which are set by the source data. Fixed seeds support computational reproducibility but do not supply inferential validity.

D Pipeline overview

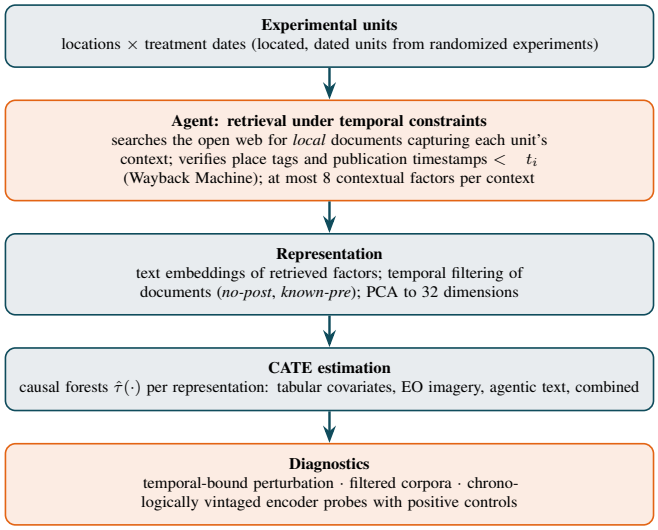


Figure 5: **The agentic pipeline.** Located, dated experimental units pass through temporally constrained agentic retrieval, temporal document filtering and embedding, causal-forest CATE estimation per representation, and the three verification layers. Orange stages are the components that enforce and verify causal admissibility.